应用：

$$\begin{aligned}z_1z_2&=rs(\cos\alpha+\mathrm{i}\sin\alpha)(\cos\beta+\mathrm{i}\sin\beta)\\&=rs[(\cos\alpha\cos\beta-\sin\alpha\sin\beta)+\mathrm{i}(\sin\alpha\cos\beta+\cos\alpha\sin\beta)]\\&=rs[\cos(\alpha+\beta)+\mathrm{i}\sin(\alpha+\beta)],\end{aligned}$$

乘积公式得证.

现在设 $z_2\neq0$（从而 $s\neq0$），用乘积公式计算复数 z_2 与 $\frac{r}{s}[\cos(\alpha-\beta)+\mathrm{i}\sin(\alpha-\beta)]$的乘积，就得到

$$\begin{aligned}&z_2\cdot\frac{r}{s}[\cos(\alpha-\beta)+\mathrm{i}\sin(\alpha-\beta)]\\=&s\cdot\frac{r}{s}[\cos(\beta+\alpha-\beta)+\mathrm{i}\sin(\beta+\alpha-\beta)]\\=&r(\cos\alpha+\mathrm{i}\sin\alpha)=z_1,\end{aligned}$$

再把等式两边同除以 z_2，就得到所求的除法公式.

例 3 计算，并用复数的代数形式表示计算结果：

(1) $\sqrt{3}\left(\cos\frac{\pi}{12}+\mathrm{i}\sin\frac{\pi}{12}\right)\cdot\sqrt{2}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)$；

(2) $\dfrac{4\left(\cos\frac{4\pi}{3}+\mathrm{i}\sin\frac{4\pi}{3}\right)}{2\left(\cos\frac{5\pi}{6}+\mathrm{i}\sin\frac{5\pi}{6}\right)}$.

解 (1) $\sqrt{3}\left(\cos\frac{\pi}{12}+\mathrm{i}\sin\frac{\pi}{12}\right)\cdot\sqrt{2}\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)$

$$\begin{aligned}&=\sqrt{3}\times\sqrt{2}\left[\cos\left(\frac{\pi}{12}+\frac{\pi}{4}\right)+\mathrm{i}\sin\left(\frac{\pi}{12}+\frac{\pi}{4}\right)\right]\\&=\sqrt{6}\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)\\&=\sqrt{6}\left(\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}\right)\\&=\frac{\sqrt{6}}{2}+\frac{3\sqrt{2}}{2}\mathrm{i}.\end{aligned}$$

(2) $\dfrac{4\left(\cos\frac{4\pi}{3}+\mathrm{i}\sin\frac{4\pi}{3}\right)}{2\left(\cos\frac{5\pi}{6}+\mathrm{i}\sin\frac{5\pi}{6}\right)}$

$$=2\left[\cos\left(\frac{4\pi}{3}-\frac{5\pi}{6}\right)+\mathrm{i}\sin\left(\frac{4\pi}{3}-\frac{5\pi}{6}\right)\right]$$